

Introduction to Microeconomics

Lecture 2—Comparative Advantage

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Years
Anniversary

ECON101
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Comparative Advantage: The Basis of Exchange

Why do people exchange goods and services in the first place? Why not just produce our own food, cars, clothing, shelter, and the like? The answer is that we can all have more of every good and service if we specialize in the activities at which we are **relatively** most efficient.

Comparative Advantage: The Basis of Exchange

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How much would it cost to produce a chicken sandwich from scratch?

According to this YouTuber^a, it took him 6 months of his life and \$1500 to make a sandwich from scratch! [▶ Link](#)

⚠ Viewer discretion advised! The video shows the murder of two chicken. 😬

^aYou should always be careful with statements said by social media creators. Some of them are knowledgeable and produce good content, but many of them are very ignorant people (that is why they became social media creators in the first place) whose only incentives are to attract the largest number of views by creating click-bait trash content. Use your critical thinking skills before believing what you hear and see in social media!

Akmaral and Sabina

	Painting	Roofing
Akmaral	300 hours	400 hours
Sabina	200 hours	100 hours

- Sabina may say that her doing both jobs is unfair. A more telling objection is that it is **inefficient**.

Akmaral and Sabina

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- This means that Akmaral has a **lower opportunity cost** of painting than Sabina does.

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- Akmaral: the opportunity cost of painting a house is 0.75 roofing jobs.

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- Sabina: the opportunity cost of painting a house is 2 roofing jobs.

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- Akmaral: the opportunity cost of painting a house is 0.75 roofing jobs.
- Sabina: the opportunity cost of painting a house is 2 roofing jobs.
- Akmaral has a comparative advantage in painting, and it makes sense for her to do both painting jobs, leaving the two roofing jobs to Sabina.

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- Akmaral: the opportunity cost of painting a house is 0.75 roofing jobs.
- Sabina: the opportunity cost of painting a house is 2 roofing jobs.
- Akmaral has a comparative advantage in painting, and it makes sense for her to do both painting jobs, leaving the two roofing jobs to Sabina.
- If each performed both the painting and roofing tasks, the total time spent would be 700 hours for Akmaral and 300 hours for Sabina. By **specializing**, the total work time falls to 600 hours for Akmaral and 200 hours for Sabina, saving 100 hours each.

Rule

To compute the opportunity cost of good A in units of good B :

$OC(A \text{ in } B) = \text{Cost of 1 unit of } A \text{ in terms of units of } B \text{ given up}$

$$\begin{aligned} &= 1 \text{ unit of } A \left(\frac{\text{Cost of } x \text{ units in resource units}}{x \text{ units of } A} \right) \left(\frac{y \text{ units of } B}{\text{Cost of } y \text{ units in resource units}} \right) \\ &= \frac{y}{x}. \end{aligned}$$

Always write units explicitly and cancel them. If units cancel correctly, the calculation is correct.

Akmaral and Sabina: Data (hours per job)

	Painting	Roofing
Akmaral	300	400
Sabina	200	100

Opportunity cost of one painting job (in roofing jobs)

Akmaral:

$$OC_A(P \text{ in } R) = (1P) \left(\frac{300 \text{ hours}}{1P} \right) \left(\frac{1R}{400 \text{ hours}} \right) = 0.75 R.$$

Sabina:

$$OC_S(P \text{ in } R) = (1P) \left(\frac{200 \text{ hours}}{1P} \right) \left(\frac{1R}{100 \text{ hours}} \right) = 2 R.$$

Conclusion

Akmaral has a **lower opportunity cost** of painting than Sabina and therefore a **comparative advantage** in painting. Sabina has a comparative advantage in roofing.

The principle of Comparative Advantage

Everyone does best when each person (or country) concentrates on the activities in which he or she is relatively most efficient.

⚠ Caution!

Pay close attention to the form in which the productivity information is provided. Your goal in each case is to find each person's opportunity cost of producing the good in question.

Sezim and Nazima

Sezim can edit a short video in 10 minutes and write a social media post in 4 minutes. Nazima can edit a short video in 6 minutes and write a social media post in 3 minutes. Which statement below is true?

- ① Sezim has no comparative advantage.
- ② Nazima should do both tasks because she has an absolute advantage in both.
- ③ Sezim has a comparative advantage in writing posts and Nazima has a comparative advantage in video editing.
- ④ Sezim has a comparative advantage in video editing and Nazima has a comparative advantage in writing posts.
- ⑤ None of the above statements is true.

Sezim and Nazima (Minutes per Task)

Sezim: 10 min per video, 4 min per post.

Nazima: 6 min per video, 3 min per post.

Opportunity cost of one video (in posts)

Sezim:

$$OC_S(V \text{ in } P) = (1V) \left(\frac{10 \text{ min}}{1V} \right) \left(\frac{1P}{4 \text{ min}} \right) = 2.5P.$$

Nazima:

$$OC_N(V \text{ in } P) = (1V) \left(\frac{6 \text{ min}}{1V} \right) \left(\frac{1P}{3 \text{ min}} \right) = 2P.$$

Conclusion

Nazima has a **lower opportunity cost** of producing videos than Sezim and therefore a **comparative advantage** in video editing.

Sezim and Nazima

Sezim can edit 10 short videos per hour or write 4 social media posts per hour. Nazima can edit 6 short videos per hour or write 3 social media posts per hour. Which statement below is true?

- ① Sezim has no comparative advantage.
- ② Nazima should do both tasks because she has an absolute advantage in both.
- ③ Sezim has a comparative advantage in writing posts and Nazima has a comparative advantage in video editing.
- ④ Sezim has a comparative advantage in video editing and Nazima has a comparative advantage in writing posts.
- ⑤ None of the above statements is true.

Sezim and Nazima (tasks per hour)

Sezim: 10 videos/hour, 4 posts/hour.

Nazima: 6 videos/hour, 3 posts/hour.

Opportunity cost of one video (in posts)

Sezim:

$$OC_S(V \text{ in } P) = (1V) \left(\frac{1 \text{ hour}}{10V} \right) \left(\frac{4P}{1 \text{ hour}} \right) = 0.4P.$$

Nazima:

$$OC_N(V \text{ in } P) = (1V) \left(\frac{1 \text{ hour}}{6V} \right) \left(\frac{3P}{1 \text{ hour}} \right) = 0.5P.$$

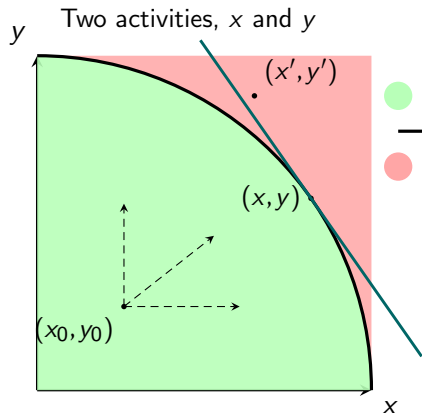
Conclusion

Sezim has a **lower opportunity cost** of producing videos than Nazima and therefore a **comparative advantage** in video editing.

The production possibilities frontier (PPF)

Production Possibilities Frontier

It is the boundary between those combinations of goods and services that can be produced, given current resources, and those that cannot.



- feasible but inefficient
- feasible & efficient
- infeasible

$|\text{slope}| = \text{C.O. of 1 more unit of } x \text{ in terms of units of } y \text{ lost.}$

$\left| \frac{1}{\text{slope}} \right| = \text{C.O. of 1 more } y \text{ in terms of units of } x \text{ lost.}$

Ali on a Remote Farm

Ali has ended up alone on a remote Stardew Valley–style farm, completely isolated from the rest of the world. He must divide his time between building shelter and producing food. In one week, Ali can produce either 6 m^2 /week of shelter or 12 kg/week of food. Describe Ali's PPF.

Opportunity cost of food

$$OC_A(f \text{ in } s) = (1f) \left(\frac{1 \text{ week}}{12f} \right) \left(\frac{6s}{1 \text{ week}} \right) = 0.5s.$$

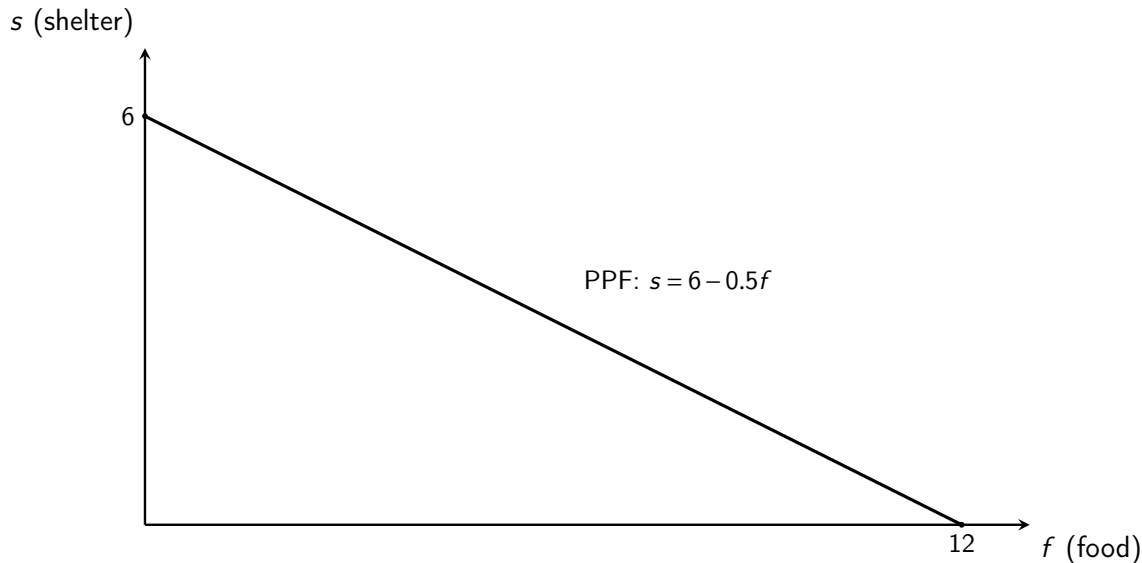
PPF

Ali's PPF is linear and given by

$$s = 6 - 0.5f,$$

with intercepts $(f, s) = (12, 0)$ and $(0, 6)$.

Ali's Production Possibility Frontier



Medina on Her Own Plot

Medina is also isolated on a nearby plot of land. If she devotes all her time to shelter, she can build 4 m^2 /week. If she devotes all her time to food production, she can produce 4 kg/week of food. Describe Medina's PPF.

Opportunity cost of food

$$OC_M(f \text{ in } s) = (1f) \left(\frac{1 \text{ week}}{4f} \right) \left(\frac{4s}{1 \text{ week}} \right) = 1s.$$

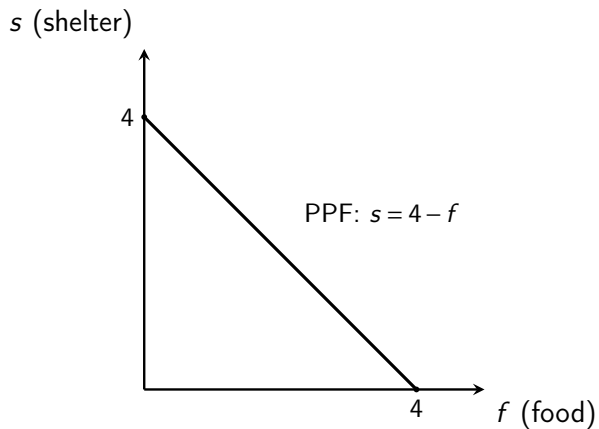
PPF

Medina's PPF is linear and given by

$$s = 4 - f,$$

with intercepts $(f, s) = (4, 0)$ and $(0, 4)$.

Medina's Production Possibility Frontier



PPF in a Two-Person Economy

Nuray, who naturally takes charge, convinces Ali and Medina to coordinate their efforts. Even though they remain isolated from the rest of the world, they now plan production jointly. In the examples above, Ali has a comparative advantage in producing food, while Medina has a comparative advantage in producing shelter.

Ali and Medina under Nuray's Organization

Ali can produce 6 m^2 /week of shelter or 12 kg/week of food. Medina can produce 4 m^2 /week of shelter or 4 kg/week of food.

Describe the economy's PPF.

PPF in a Two-person farm: Solution

Individual technologies

In one week:

- Ali can produce either 12 f or 6 s .
- Medina can produce either 4 f or 4 s .

Comparative advantage

Ali has a lower opportunity cost of producing food; Medina has a lower opportunity cost of producing shelter.

Joint production possibilities

- If both specialize in shelter: $(0,6) + (0,4) = (0,10)$.
- If both specialize in food: $(12,0) + (4,0) = (16,0)$
- If both specialize according to comparative advantage:

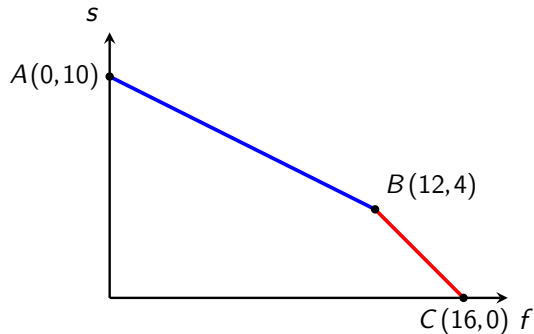
$$(12,0) + (0,4) = (12,4)$$

Joint PPF

The economy's PPF is piecewise linear with a kink at $(12,4)$, connecting

$$(0,10) \rightarrow (12,4) \rightarrow (16,0).$$

Joint Production Possibility Frontier



A: both specialize in shelter.

B: Ali specializes in food, Medina in shelter (comparative advantage).

C: both specialize in food.

— Segment A–B: Ali partially food & shelter, Medina only shelter; $OC(f \text{ in } s) = 0.5$.

— Segment B–C: Ali only food, Medina partially food & shelter; $OC(f \text{ in } s) = 1$.

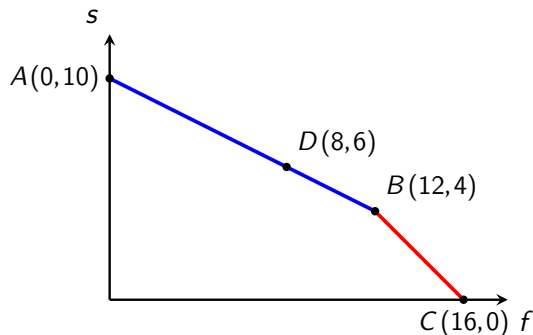
$$s(f) = \begin{cases} 10 - \frac{1}{2}f, & 0 \leq f \leq 12, \\ 16 - f, & 12 \leq f \leq 16. \end{cases}$$

Production Assignment on the Farm

Under Nuray's direction, Ali and Medina decide to jointly consume $6 \text{ m}^2/\text{week}$ of shelter and 8 kg/week of food.

How should Nuray assign the production tasks between Ali and Medina?

Joint Production Possibility Frontier: Example of joint production



A: both specialize in shelter.

B: Ali specializes in food, Medina in shelter (comparative advantage).

C: both specialize in food.

— Segment A-B: Ali partially food & shelter, Medina only shelter; $OC(f \text{ in } s) = 0.5$.

— Segment B-C: Ali only food, Medina partially food & shelter; $OC(f \text{ in } s) = 1$.

$$s(f) = \begin{cases} 10 - \frac{1}{2}f, & 0 \leq f \leq 12, \\ 16 - f, & 12 \leq f \leq 16. \end{cases}$$

At point $D = (8,6)$: Medina produces $(0,4)$ and Ali produces $(8,2)$ jointly.

The Principle of Increasing Opportunity Cost

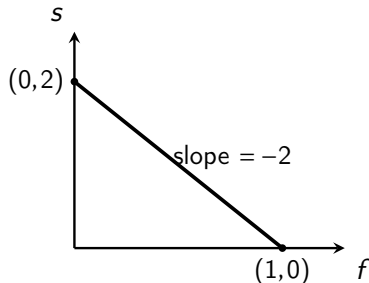
In expanding the production of any good, first employ those workers whose opportunity cost of producing that good is lowest, and only afterward turn to workers with higher opportunity costs.

Nuray Joins the Farm

Eventually, Nuray herself joins the production process. She can produce 2 m^2 /week of shelter or 1 kg/week of food.

Start by describing Nuray's individual PPF. Then describe the PPF of the three-person economy under her coordination.

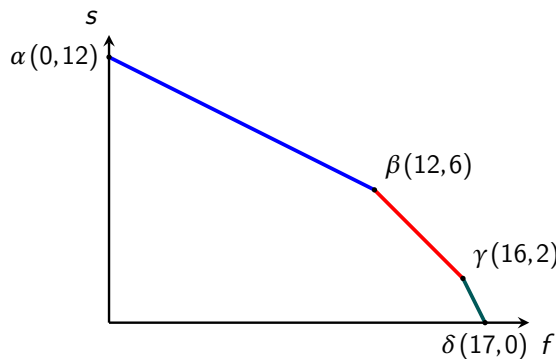
If they jointly decide to produce 14 kg/week of food and 4 m^2 /week of shelter, how should Nuray allocate each person's work time?



Agent	OC _{<i>i</i>} (f in s)
A (Ali)	0.5
M (Medina)	1
N (Nuray)	2

$A < M < N$ (in increasing OC of food).

Three-Person Joint Production Possibility Frontier



α : all specialize in shelter.

β : A specializes in food; M, N in shelter.

γ : A, M specialize in food; N in shelter.

δ : all specialize in food.

— $\alpha\beta$: switch A to food; $OC(f \text{ in } s) = 0.5$.

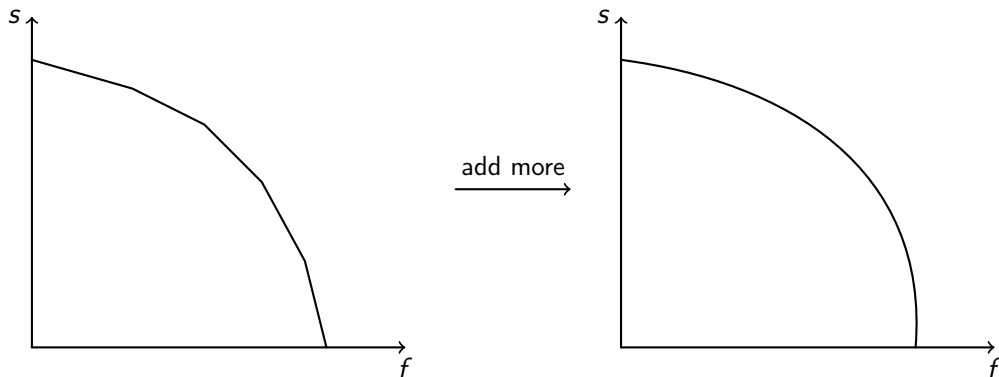
— $\beta\gamma$: switch M to food; $OC(f \text{ in } s) = 1$.

— $\gamma\delta$: switch N to food; $OC(f \text{ in } s) = 2$.

$$s(f) = \begin{cases} 12 - \frac{1}{2}f, & 0 \leq f \leq 12, \\ 18 - f, & 12 \leq f \leq 16, \\ 34 - 2f, & 16 \leq f \leq 17. \end{cases}$$

The production possibilities frontier with many workers

Intuition behind its shape: produce the initial units of clothing using the resources that are relatively most efficient at clothing production, and only then turn to those that are relatively less efficient at clothing production.



Another example of a bowed out (strictly concave) PPF

A Production Possibilities Frontier with Labor as the Only Input

Suppose that there are two goods, x and y , which require labor as an input of production. Let l_x and l_y denote the amounts of labor used in the production of x and y , respectively. Suppose that there are 100 hours of labor available.

Assume also that the production processes of x and y are given by

$$x = \sqrt{l_x}, \quad y = \sqrt{l_y}.$$

Describe, mathematically and graphically, the production possibilities frontier (PPF).

Curved PPF from a Time (Resource) Constraint

Resource constraint (time):

$$100 = \ell_X + \ell_Y, \quad \ell_X, \ell_Y \geq 0.$$

Production technologies:

$$x = \sqrt{\ell_X}, \quad y = \sqrt{\ell_Y}.$$

Equivalently,

$$\ell_X = x^2, \quad \ell_Y = y^2.$$

Substitute into the time constraint:

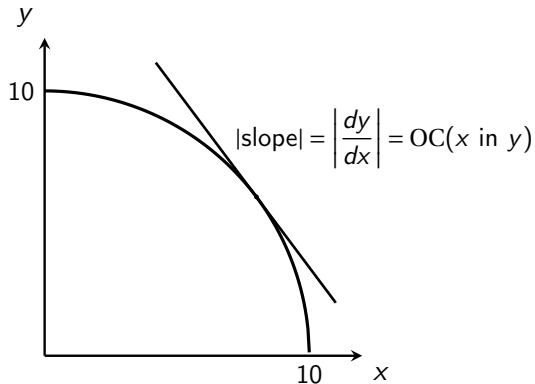
$$100 = x^2 + y^2, \quad x \geq 0, y \geq 0.$$

Solving for y (the PPF in (x, y) -space):

$$y = \sqrt{100 - x^2}, \quad 0 \leq x \leq 10.$$

Opportunity cost interpretation. Along the frontier,

$$\left| \frac{dy}{dx} \right| = \text{units of } y \text{ sacrificed for one more unit of } x = \text{OC}(x \text{ in } y).$$



A second isolated farm

Erika, Bagdat, and Sultan

On a different isolated farm, three individuals decide to coordinate production.

Weekly production possibilities:

- Erika can produce 3 m^2 /week of shelter *or* 9 kg/week of food.
- Bagdat can produce 8 m^2 /week of shelter *or* 4 kg/week of food.
- Sultan can produce 6 m^2 /week of shelter *or* 6 kg/week of food.

Sultan decides to organize the three of them and becomes the boss.

Describe the PPF of this three-person economy.

Each agent has a linear individual PPF between full shelter and full food.

Erika: $(f, s) = (9, 0), (0, 3)$,

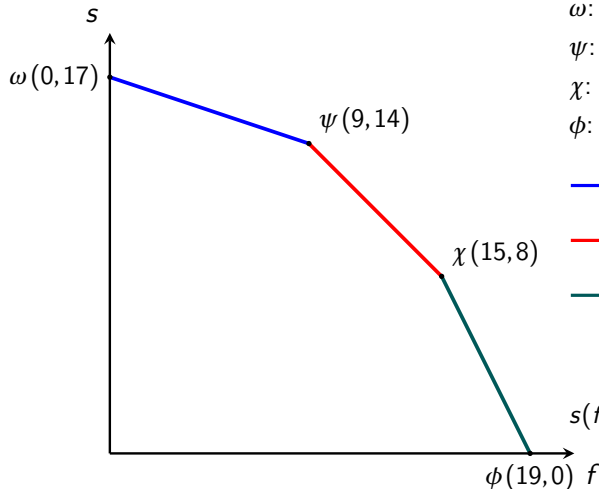
Bagdat: $(f, s) = (4, 0), (0, 8)$,

Sultan: $(f, s) = (6, 0), (0, 6)$.

Agent	$OC_i(f \text{ in } s)$
Erika	$\frac{1}{3}$
Sultan	1
Bagdat	2

$$E < S < B$$

Joint Production Possibility Frontier



ω : all specialize in shelter.

ψ : Erika in food; Sultan, Bagdat in shelter.

χ : Erika, Sultan in food; Bagdat in shelter.

ϕ : all specialize in food.

— Segment ω – ψ : $OC(f \text{ in } s) = \frac{1}{3}$.

— Segment ψ – χ : $OC(f \text{ in } s) = 1$.

— Segment χ – ϕ : $OC(f \text{ in } s) = 2$.

$$s(f) = \begin{cases} 17 - \frac{1}{3}f, & 0 \leq f \leq 9, \\ 23 - f, & 9 \leq f \leq 15, \\ 38 - 2f, & 15 \leq f \leq 19. \end{cases}$$

The two farms get organized—From isolation to interaction

Two farms discover each other

Until now, the two farms have been completely isolated.

Suddenly, they discover that the other farm exists.

Production technologies do not change. What changes is that they can now *trade* with each other.

Organizing into firms

As soon as trade becomes possible, something remarkable happens:

- Nuray organizes her farm and calls it **Nonchalant Empire™**.
- Sultan organizes his farm and calls it **Superior Farm™**.

Titles are assigned. Logos are imagined. PowerPoint decks are created.

Economically speaking, these are still just groups of workers with production technologies.

Nonchalant Empire technology recap

Nonchalant Empire consists of three workers:

- Ali can produce 6 m^2 of shelter or 12 kg of food.
- Medina can produce 4 m^2 of shelter or 4 kg of food.
- Nuray can produce 2 m^2 of shelter or 1 kg of food.

Compute Nonchalant Empire's opportunity costs and identify its comparative advantage.

Superior Farm technology recap

Superior Farm consists of three workers:

- Erika can produce 3 m^2 of shelter or 9 kg of food.
- Bagdat can produce 8 m^2 of shelter or 4 kg of food.
- Sultan can produce 6 m^2 of shelter or 6 kg of food.

Compute Superior Farm's opportunity costs and identify its comparative advantage.

Since the PPFs are piecewise linear, the opportunity cost depends on the segment.

Nonchalant Empire

Joint PPF:

$$s^{NE}(f) = \begin{cases} 12 - \frac{1}{2}f, & 0 \leq f \leq 12, \\ 18 - f, & 12 \leq f \leq 16, \\ 34 - 2f, & 16 \leq f \leq 17. \end{cases}$$

Opportunity cost of food:

$$OC^{NE}(f \text{ in } s) = \begin{cases} \frac{1}{2}, & 0 \leq f \leq 12, \\ 1, & 12 \leq f \leq 16, \\ 2, & 16 \leq f \leq 17. \end{cases}$$

$$\min OC^{NE} = \frac{1}{2}.$$

Superior Farm

Joint PPF:

$$s^{SF}(f) = \begin{cases} 17 - \frac{1}{3}f, & 0 \leq f \leq 9, \\ 23 - f, & 9 \leq f \leq 15, \\ 38 - 2f, & 15 \leq f \leq 19. \end{cases}$$

Opportunity cost of food:

$$OC^{SF}(f \text{ in } s) = \begin{cases} \frac{1}{3}, & 0 \leq f \leq 9, \\ 1, & 9 \leq f \leq 15, \\ 2, & 15 \leq f \leq 19. \end{cases}$$

$$\min OC^{SF} = \frac{1}{3}.$$

$$\frac{1}{3} < \frac{1}{2}$$

$\Rightarrow$

Superior Farm has comparative advantage in food (at the margin).

When is trade beneficial?

Terms of trade

Let p denote the **terms of trade** or more colloquially, the **price** of food in units of shelter:

$$p = \frac{m^2 \text{ of shelter}}{\text{kg of food}}.$$

Trade is mutually beneficial if p lies strictly between the opportunity costs of producing food in the two corporations:

$$OC^{SF}(\text{food}) < p < OC^{NE}(\text{food}).$$

Find such a value of p .

Range of mutually beneficial prices

From the joint PPFs:

$$OC^{SF}(\text{food}) = \frac{1}{3}, \quad OC^{NE}(\text{food}) = \frac{1}{2}.$$

Hence, any price satisfying

$$\frac{1}{3} < p < \frac{1}{2}$$

makes trade strictly beneficial for both corporations.

Predicting behavior

Suppose a price p satisfying the inequality has been chosen.

- Which corporation specializes relatively more in food?
- Which corporation specializes relatively more in shelter?
- Why does this follow from comparative advantage?

Predicting behavior

Fix any p with $\frac{1}{3} < p < \frac{1}{2}$.

- **SF specializes relatively more in food** (and exports food).
- **NE specializes relatively more in shelter** (and exports shelter).
- **Reason (comparative advantage):**
 $OC^{SF}(\text{food}) = \frac{1}{3} < p \Rightarrow$ SF can produce food at lower cost than the trading price;
 $p < OC^{NE}(\text{food}) = \frac{1}{2} \Rightarrow$ NE finds producing food relatively expensive and shifts toward shelter.

A concrete trade scenario

Choose a price $p =$ _____.

Suppose:

- Nonchalant Empire exports _____ m^2 of shelter.
- Superior Farm exports _____ kg of food.

Verify:

- Each firm can produce its export bundle.
- After trade, each firm can consume a bundle that was infeasible under autarky (no-trade).

Conclude: trade is beneficial.

The trade line

Fix a price p measured as

$$p = \frac{m^2 \text{ of shelter}}{\text{kg of food}}.$$

Starting from any production bundle (f, s) , trading at price p means:

Let $\Delta f = (f' - f)$ and $\Delta s = (s' - s)$.

- if the farm *buys* $\Delta f > 0$ units of food, it must *pay* $p\Delta f$ units of shelter;
- if the farm *sells* $\Delta f > 0$ units of food, it *receives* $p\Delta f$ units of shelter.

So all bundles that can be reached by trading from (f, s) satisfy

$$\Delta s = -p\Delta f \Rightarrow s' - s = -p(f' - f) \Rightarrow s' = s - p(f' - f) \Rightarrow s' + pf' = s + pf.$$

This is a straight line through (f, s) with slope $-p$.

Trade at $p = 0.4$: predicted production points

Apply to $p = 0.4$ using opportunity costs

At $p = 0.4$ compare the price to the (segment) opportunity costs of producing food.

Nonchalant Empire (NE):

$$OC^{NE}(\text{food}) \in \left\{ \frac{1}{2}, 1, 2 \right\}, \quad 0.4 < \frac{1}{2} < 1 < 2.$$

Since p is *below the smallest* opportunity cost of food, NE does not produce food at the margin and chooses the max-shelter vertex:

$$\text{NE chooses } (f, s) = (0, 12).$$

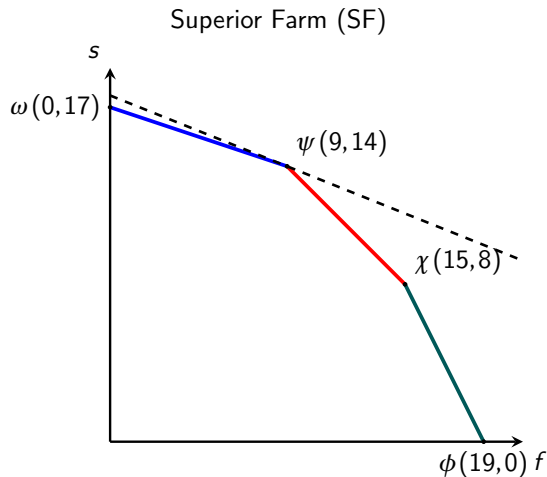
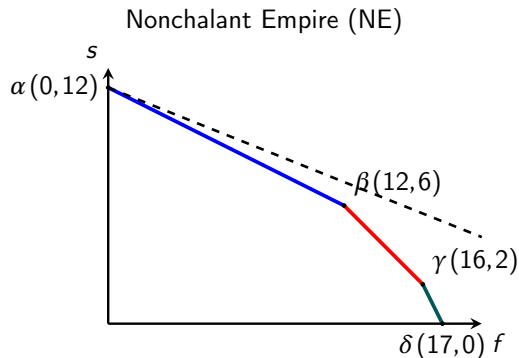
Superior Farm (SF):

$$OC^{SF}(\text{food}) \in \left\{ \frac{1}{3}, 1, 2 \right\}, \quad \frac{1}{3} < 0.4 < 1 < 2.$$

Here p lies between the first and second opportunity costs, so SF produces food along the first segment and stops at the kink where the marginal OC jumps above p :

$$\text{SF chooses } (f, s) = (9, 14).$$

Trade at $p = 0.4$: Graphical verification



Graphical procedure: draw a line with slope $-p$ and slide it outward until it first touches the joint PPF; the contact point (a vertex or a kink) is the optimal production choice.